

Quark Compositions

AQA A-Level Physics Specification (2026)

Name: _____

Date: _____

1. What is the quark composition of a meson?A. qqq B. $q\bar{q}q$ C. $q\bar{q}$

D. Not made up of quarks

2. What is the quark composition of a baryon?A. qqq B. $q\bar{q}q$ C. $q\bar{q}$

D. Not made up of quarks

3. What is the quark composition of a lepton?A. qqq B. $q\bar{q}q$ C. $q\bar{q}$

D. Not made up of quarks

4. What is the quark composition of an antibaryon?A. qqq B. $q\bar{q}q$ C. $q\bar{q}$

D. Not made up of quarks

5. What is the quark composition of a proton?A. uud B. udd C. $u\bar{d}$

D. Not made up of quarks

6. What is the quark composition of a neutron?A. uud B. udd C. $u\bar{d}$

D. Not made up of quarks

7. What is the quark composition of a negative kaon?A. $u\bar{d}$ B. $s\bar{u}$ C. $s\bar{d}$

D. Not made up of quarks

8. What is the quark composition of a neutral kaon?A. $\bar{u}s$ B. $u\bar{s}$ C. $d\bar{s}$

D. Not made up of quarks

9. What is the quark composition of a positive kaon?A. $d\bar{s}$ B. $\bar{d}s$ C. $u\bar{s}$

D. Not made up of quarks

10. What is the quark composition of an electron?

- A. uds
- B. $\bar{d}s$
- C. $d\bar{s}$
- D. Not made up of quarks

11. What is the quark composition of a positive pion?

- A. $u\bar{d}$
- B. $d\bar{u}$
- C. $u\bar{u} / d\bar{d}$
- D. Not made up of quarks

12. What is the quark composition of a neutral pion?

- A. $u\bar{d}$
- B. $d\bar{u}$
- C. $u\bar{u} / d\bar{d}$
- D. Not made up of quarks

13. What is the quark composition of a negative pion?

- A. $u\bar{d}$
- B. $d\bar{u}$
- C. $u\bar{u} / d\bar{d}$
- D. Not made up of quarks

14. What is the quark composition of a muon?

- A. $s\bar{s}$
- B. $u\bar{d}$
- C. $d\bar{u}$
- D. Not made up of quarks

15. What is the quark composition of an antiproton?

- A. $\bar{u}\bar{d}\bar{d}$
- B. $u\bar{d}$
- C. $\bar{u}\bar{u}\bar{d}$
- D. Not made up of quarks

16. What is the quark composition of an antineutron?

- A. $\bar{u}\bar{d}\bar{d}$
- B. $u\bar{d}$
- C. $\bar{u}\bar{u}\bar{d}$
- D. Not made up of quarks

17. What is the quark composition of a positron?

- A. $\bar{u}\bar{d}\bar{d}$
- B. $u\bar{d}$
- C. $\bar{u}\bar{u}\bar{d}$
- D. Not made up of quarks

18. What is the quark composition of an antimuon?

- A. $\bar{u}\bar{d}\bar{d}$
- B. $u\bar{d}$
- C. $\bar{u}\bar{u}\bar{d}$
- D. Not made up of quarks

19. What is the quark composition of a neutrino?

- A. $\bar{u}\bar{d}\bar{d}$
- B. $u\bar{d}$
- C. $\bar{u}\bar{u}\bar{d}$
- D. Not made up of quarks

Answer Key

1. C
2. A
3. D
4. B
5. A
6. B
7. B

8. C
9. C
10. D
11. A
12. C
13. B
14. D

15. C
16. A
17. D
18. D
19. D